Associations between Use of Targeted Therapies, Hospitalizations and Emergency Department Visits Among Patients with Atopic Dermatitis: A Retrospective Cohort Study

Richard Ta, Pharm.D, M.S.
The CHOICE Institute, Department of Pharmacy, University of Washington

Jong Min Jung, M.S., M.A.
Department of Economics, University of Washington

Kyu Eun Lee, Ph.D.
The CHOICE Institute, Department of Pharmacy, University of Washington

Jing Li, Ph.D.
The CHOICE Institute, Department of Pharmacy, University of Washington

Corresponding Author:
Jing Li
1959 NE Pacific Street
Health Sciences Building
Seattle, WA 98195-7360
University of Washington
jli0321@uw.edu

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Abstract

Background: Atopic dermatitis (AD) is the most common inflammatory dermatologic condition. Despite approvals of numerous targeted therapies, no studies have examined their real-world relationship with acute healthcare utilizations and healthcare spending for AD patients.

Objective: To assess the association between use of targeted therapy and the probability of hospitalizations and emergency department (ED) visits and healthcare spending among adult patients with AD in the United States.

Methods: This retrospective cohort study used MarketScan commercial insurance claims data from January 1, 2012, through December 31, 2023, with analyses conducted between November 1, 2023, and August 1, 2024. Adult patients (≥18 years) with a diagnosis of AD who had at least one prescription claim for a targeted AD therapy (upadacitinib, abrocitinib, tralokinumab, baricitinib, dupilumab) were included. The primary outcomes were any hospitalizations and any ED visits in a given year. The secondary outcomes were annual total healthcare spending and spending by type of service (drug, inpatient, and outpatient), and patient out-of-pocket costs. A Difference-in-Difference (DiD) design was employed to compare changes in outcomes between AD patients who received targeted therapies and those who did not receive a prescription for these therapies.

Results: Among 157,966 (mean [SD] age 41.6 [13] years; 55,077 [35%] male) individuals with AD, 9,320 (6%) received prescriptions for a targeted therapy, among whom 99% received dupilumab. The DiD results showed that targeted therapy use was associated with a decrease of 1.6 percentage points (ppts) (95% CI: -0.021 to -0.010; p-value < 0.01) in the probability of any hospitalization and a 1.7 percentage points (95% CI: -0.025 to -0.008; p-value < 0.01) decrease in the probability of any ED visits. Targeted therapy was associated with an increase in overall healthcare spending ($24,901; 95% CI: $24,482 to $25,320) which was entirely due to increases in drug spending ($26,061; 95% CI: $25,735 to $26,388), along with decreases in inpatient and outpatient spending. It was also associated with a $95 increase in annual patient out-of-pocket spending (95% CI: $83 to $108).

Conclusions: Targeted AD therapy was associated with fewer all-cause hospitalization and ED visits and lower inpatient and outpatient spending, although higher drug spending resulted in greater total healthcare spending.

Plain Language Summary

Atopic dermatitis (AD) is a common long-term skin disease that can be severe for some adults. We compared adults who started new drugs for AD with adults who did not use them. Adults who started these drugs had fewer hospital stays and fewer emergency department visits. Their total health care costs were higher as the drugs were expensive. However, patients paid only slightly more out of pocket.

Implications for Managed Care Pharmacy

This study shows that targeted AD therapies reduced hospitalizations and ED visits despite higher drug costs. These findings support benefit designs that prioritize access to effective biologics and JAK inhibitors to offset costly acute care. Results may inform value-based coverage, step therapy refinement, and formulary placement favoring high-value agents. Improving access to targeted therapies may also reduce inequities by limiting preventable hospital-based care among vulnerable populations.

Introduction

Atopic dermatitis (AD) is the most prevalent inflammatory dermatologic disorder, affecting approximately 2% to 10% of adults and 10% to 30% of children.[1] Its common manifestations include pruritus, dry skin, rashes, and erythema.[1] If left untreated, AD significantly impacts patients' quality of life, leading to sleep disturbances, decreased productivity at school or work, and adverse effects on mental health.[2] The chronic nature and increasing prevalence of AD contribute to a substantial economic burden in the United States (US). In 2015, the estimated annual cost of AD in the US amounted to approximately $5.3 billion.[3] Furthermore, a study revealed a notable rise in AD-related total emergency department costs from $265 million in 2006 to $642 million in 2012.[4]

Conventional immunosuppressants were the main systemic treatment for moderate-to-severe AD until 2017.[5] Since then, the treatment landscape has changed substantially with the approval of systemic targeted therapies, including biologics such as dupilumab and tralokinumab as well as oral Janus Kinase inhibitors such as abrocitinib, upadacitinib, and baricitinib.[6] These therapies target inflammatory pathways involved in AD and have been shown to significantly improve disease severity, skin clearance, and quality of life.[7,8]

Hospitalizations and ED visits are critical outcomes for patients, clinicians, and payers as they represent high-intensity and high-cost healthcare use. Adults who are diagnosed with AD may need these visits when their condition worsens or becomes difficult to manage, including sudden flare-ups, skin infections, complications from other health problems, or when their current treatment is no longer effective or causes side effects. By improving disease control and alleviating symptoms, these targeted therapies may significantly reduce the need for acute and urgent care. However, evidence supporting these therapies has relied primarily on clinical severity and symptom endpoints from randomized trials, leaving very limited real-world evidence, largely descriptive in nature, on whether these therapies are associated with lower hospitalizations, ED visits, and healthcare spending relative to non-targeted-therapy care.[24,25]

In this study, we address this gap by examining the association between targeted therapy initiation and all-cause hospitalization, all-cause ED visits, and healthcare spending among commercially insured adults diagnosed with AD. We used a Difference-in-Differences (DiD) design to compare changes in these outcomes before and after the index date between individuals who received targeted therapy and those who did not.

Methods

Data and Study Period

This retrospective cohort study used data from the Merative® MarketScan® commercial database, specifically the Commercial Claims and Encounters (CCAE) database, from January 1, 2012, through December 31, 2023.[9] The study received exempt status by the institutional review board of the University of Washington. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines.

Study Population and Exposure

We identified adults with AD during the patient enrollment period from January 1, 2017, through December 31, 2022. AD was defined as having at least one inpatient claim or at least two outpatient claims with the International Classification of Diseases (, Tenth Revision (ICD-10-CM) diagnosis codes beginning with L20 in any billing position (Supplementary Table 1), consistent with prior claims-based studies of adult AD.[10,11] Patients were included if they were 18 years or older and continuously enrolled in medical and pharmacy benefits for at least 12 months immediately before and after the index date (defined below).

The targeted therapy group included adults with AD who received abrocitinib, dupilumab, tralokinumab, baricitinib and upadacitinib during the patient enrollment period.[12] Targeted therapies were identified using National Drug Codes (Supplementary Table 2) and the index date was the date of the first observed targeted-therapy prescription claim. To identify new users of targeted therapy, patients were required to have no targeted-therapy prescription claim during the 12-month baseline washout period before the index date, consistent with new-user designs commonly used in pharmacoepidemiologic claims studies.[13]

The comparison group included adults with AD who never received any targeted therapy during the study period (January 1, 2012-December 31, 2023) and were continuously enrolled in medical and pharmacy benefits for at least 24 months between January 1, 2016 and December 31, 2023. As these patients may still have received topical therapy, conventional systemic therapy, or supportive care, we refer to this group as the non-targeted-therapy comparison group. We did not further distinguish among patients in the comparison group based on alternative treatments, as we were not able to identify any treatments received over the counter, and the current broad definition of comparison group reflects the real-world landscape of AD population who did not receive the targeted therapy, including both untreated and those with alternative therapies, forming a meaningful comparison to our targeted therapy group. For the comparison group, the index date was 12 months after the start of continuous enrollment period. For patients in the comparison group with multiple continuous enrollment periods, we kept the first instance only.

Patients were excluded if they did not meet continuous enrollment requirements, were younger than 18 years during the continuous enrollment period, or did not meet the claims-based AD definition.

Study Outcomes

The primary outcomes of interest were two indicators for healthcare utilization : (1) any all-cause hospitalizations and (2) any all-cause ED visits. We selected all-cause hospitalization and ED visits as they capture high-intensity acute care use relevant to patients, clinicians, and payers. These outcomes were intended to measure population-level, payer-relevant acute care use after targeted therapy initiation, rather than to attribute each encounter directly to AD or treatment. Thus, the estimates do not identify whether the specific encounter was directly attributable to AD, treatment failure, treatment-related adverse events, or other conditions. We did not restrict to AD-specific or adverse-event-specific ICD codes, as acute care driven by poorly controlled AD, such as secondary skin infections, is often coded to the proximate clinical diagnosis rather than to AD itself. A code-restricted definition therefore risks excluding clinically severe, policy-relevant encounters and understating true acute care burden. Any AD-unrelated encounters captured in these outcomes are addressed by our difference-in-differences design, which accounts for fixed differences in baseline levels between groups and would only introduce bias if trends in AD-unrelated encounters diverged systematically between groups over time, which there is no clinical reason to expect.

The secondary outcomes of interest were annual all-cause total healthcare spending and spending categories including drug, inpatient, and outpatient spending. Total healthcare spending was defined as the sum of inpatient, outpatient, and drug spending. Drug spending included all prescription drug spending captured in claims and was not limited to AD-specific medications. Out-of-pocket (OOP) spending was also examined, including total OOP spending as well as drug, outpatient, and inpatient OOP spending. All spending values were winsorized at the 99th percentile to mitigate the influence of outliers, and adjusted for inflation using the medical care component of the consumer price index to reflect values in 2023 dollars.[14]

Statistical Analysis

We summarized the baseline patient characteristics using mean and standard deviation (SD) for continuous variables and percentages for categorical variables. These included the patient's sex, age, geographic region, and targeted therapy prescribed (if any) on the index date. The Charlson Comorbidity Index (CCI) was calculated using comorbidity, an R package, which uses ICD-10 diagnosis codes on medical claims collected during the 12-month pre-index period.[15-18] We assessed the differences in baseline characteristics between the two groups using the Mann-Whitney test for age and chi-square test for sex, region, and CCI.

We employed a difference-in-differences (DiD) study design to estimate the impact of targeted therapy on outcomes. The DiD design compares changes in outcomes before and after the index date between the targeted therapy group and the non-targeted therapy comparison group (See Supplementary Methods 1 for detailed model specification).

In the DiD model, the coefficient of interest was the interaction between the targeted therapy group indicator and the post-index period indicator. The model adjusted for person fixed effects to control for time-invariant individual characteristics and calendar year-month fixed effects to control for secular trends, seasonality, and changes in healthcare practice common to both groups. We additionally controlled for time-varying comorbidity burden using CCI. We did not separately measure all possible time-varying clinical factors such as dermatologic severity, immunologic conditions, or use of non-targeted AD therapies. These factors were not included because they typically change in response to treatment itself, and adjusting for them would risk removing part of the true treatment effect rather than controlling for confounding. Standard errors were clustered at the individual patient level. In addition, spending outcomes were analyzed as continuous outcomes using linear DiD models to estimate adjusted mean differences in spending. We used linear models as mean spending differences are directly interpretable for payer and managed care decision making, and additionally estimated Poisson models for spending outcomes in sensitivity analysis.

The DiD method requires the parallel trends assumption to yield plausibly causal estimates of treatment effects. The assumption posits that, in the absence of targeted therapy initiation, the targeted therapy group and non-targeted therapy group would have followed similar trends over time. We assessed the plausibility of the parallel trends assumption through visual inspection of the mean outcomes for both groups over time.

All statistical analyses were performed using R Statistical Software (v4.2.2; R Core Team 2024).

Results

Baseline Characteristics

We identified a total of 157,966 individuals diagnosed with AD in the MarketScan database between January 1, 2017, and December 31, 2022, who met the eligibility criteria. These included 148,646 non-targeted therapy comparison individuals and 9,320 individuals who received targeted therapy (Figure 2), 99% of whom were prescribed dupilumab (Table 1). The final analysis sample included 796,254 person-years, including 51,738 person-years from the treated group and 744,516 person-years from the comparison group. The distribution of person-years across all years for the treated and comparison group is reported in Supplementary Figure 1.

The mean [SD] age at baseline was 41.1 [13.6] years for the treated group and 41.6 [14.1] years for the comparison group ($p = 0.008$) (Table 1). The treated group had a higher proportion of males than the comparison group (42% vs. 34%, $p < 0.001$). The treated group had more individuals from the south, while the comparison group had more from the northeast and west ($p < 0.001$). The treated group had a lower proportion of individuals with a CCI score of 0 (46% vs. 57%) and a higher proportion with a CCI score of ≥ 3 (13% vs 10%, p-value < 0.001).

DiD Results on Primary Outcomes

The unadjusted trends in the probability of any hospitalizations (Panel A) and the probability of any ED visits (Panel B) are shown in Supplementary Figure 2. The trends in the probability of any ED (Panel B) visits were generally parallel between the targeted therapy group and non-targeted therapy comparison group prior to the index date. The trends in the probability of any hospitalization show greater variability but are broadly consistent with the parallel trends assumption. Table 2 shows the unadjusted means by group and time period for the primary outcomes and the adjusted treatment estimates using DiD regressions.

In unadjusted analyses, the proportion of subjects with any hospitalization decreased from pre-index to the post-index period in both the comparison group (from 7.0% to 6.8%) and the targeted therapy group (from 7.2% to 6.0%). The proportion of patients with any ED visit increased slightly in the comparison group (from 16.2% to 16.9%) and decreased in the targeted therapy group (from 19.0% to 17.9%) over the same period.

After controlling for covariates and fixed effects, targeted therapy initiation was associated with an additional 1.6 percentage point (95% CI: -0.021, -0.010; p-value < 0.01) decrease in the probability of any hospitalizations and additional 1.7 percentage point (95% CI: -0.025, -0.008; p-value < 0.01) decrease in the probability of any ED visits compared with the non-targeted therapy comparison group.

Secondary Outcomes

The unadjusted trends for the healthcare spending outcomes are shown in Supplementary Figure 3. Pre-index spending trends were generally similar for most of the pre-index period. However, total healthcare and drug spending increased among targeted-therapy groups in the year immediately before the index date. Therefore, spending estimates should be interpreted more cautiously in our study context.

The unadjusted means results for healthcare spending showed increases across all categories in the comparison group between the pre-index and post-index periods, including mean total spending (from $7,979 to $8,955), drug spending (from $2,156 to $2,499), outpatient spending (from $4,827 to $5,396), and inpatient spending (from $996 to $1,060). In the treated group, total and drug spending rose sharply, with total spending increasing from $11,011 to $37,935 and drug spending from $4,216 to $31,000. Outpatient spending increased slightly (from $5,691 to $5,888), while inpatient spending saw a small decline (from $1,104 to $1,047) over the same period.

The adjusted results from the DiD regressions showed that targeted therapy was associated with a $24,901 (95% CI: 24,482, 25,320; p-value < 0.01) increase in annual total spending and a $26,061 (95% CI: 25,734, 26,387; p-value < 0.01) increase in annual drug spending compared to those not on a targeted therapy. Targeted therapy was also associated with a $909 (95% CI: -1,106, -711; p-value < 0.01) decrease in outpatient spending and $251 (95% CI: -382, -119; p-value < 0.01) decrease in inpatient spending when compared to those not on a targeted therapy (Table 3). Results from Poisson regression specification (omitted) were consistent in direction and statistical significance with our primary linear DiD estimates.

Targeted therapy was also associated with a \$95 increase in annual total OOP spending (95% CI: 83, 108; $p < 0.01$), driven by higher drug and inpatient OOP costs but lower outpatient OOP costs (Supplementary Table 3). Overall, OOP spending trends mirrored those of non-OOP spending, except inpatient spending decreased and inpatient OOP costs increased only slightly.

Discussion

We conducted a retrospective cohort study using MarketScan claims data to estimate the associations between targeted therapy initiation, ED visits, hospitalizations and healthcare spending among adults diagnosed with AD. On average, targeted therapy was associated with a 1.6-percentage point decrease in the probability of any hospitalization and 1.7-percentage point decrease in the probability of any ED visits compared with the non-targeted therapy comparison group. In relative terms, these estimates correspond to approximately 23% and 10% reductions, respectively, calculated by dividing DiD estimates by the corresponding comparison group rates (i.e. 1.6 percentage points divided by 7.0% for hospitalization and 1.7 percentage points divided by 16.9% for ED visits).

Targeted therapies have been demonstrated to improve symptoms associated with AD such as skin clearance, reduction in itchiness, and decreased frequency of flares. The resulting improvement in symptom management can lead to a decreased likelihood of requiring an ED visit or hospitalization. Studies have also shown that targeted therapy can effectively manage AD symptoms which can lead to a significant improvement in a patient's quality of life by enhancing sleep patterns and reducing anxiety and depression, perhaps further reducing the likelihood of related hospitalizations and ED visits.[19-20]

Targeted therapy was further associated with a \$24,901 increase in mean total spending and \$26,061 increase in mean drug spending, while also contributing to a \$909 decrease in mean outpatient spending and \$251 decrease in mean inpatient spending. Despite a decrease in ED visits and hospitalizations, there was a significant rise in total spending, mainly driven by higher drug spending likely due to the high prices of targeted therapies. Recent research also highlighted the substantial spending of targeted therapies, revealing out of an average of \$2,381 spent on all-cause healthcare spending for adults with AD, \$1,997 (84%) was attributed to outpatient pharmacy claims, with 72% of that amount being spent on targeted therapies.[11] Compared to the control mean of the study period, targeted therapy was associated with a 280% increase in mean total spending, and a 1,123% increase in mean drug spending. Meanwhile, the relative decreases in mean outpatient and inpatient spending were 17.8% and 24%, respectively. Although these results highlight the real-world tradeoff between reductions in intensive healthcare utilizations and increases in drug spending, the overall cost-effectiveness of targeted therapies remain uncertain. Cost-utility analyses that account for quality-of-life improvements have found targeted therapies like dupilumab as cost-effective compared to standard of care, particularly for patients with more severe AD who experience greater improvement in quality of life.[21,22] This highlights that claims data alone cannot fully capture the comprehensive benefits of targeted therapy, as it overlooks factors such as indirect costs other intangible benefits. Consistent with this interpretation, a cost-of-illness study demonstrated a substantial economic burden associated with AD, primarily driven by productivity loss from work absenteeism and diminished work performance.[23]

A similar trend was observed with the initiation of targeted therapy on OOP spending, which was associated with a $95 increase in mean total OOP spending (Supplementary table 3). This increase was primarily driven by a $140 rise in mean drug OOP spending, partially offset by a $45 reduction in outpatient OOP spending. The disproportionate rise in drug spending relative to drug OOP spending suggests that cost-sharing mechanisms may help mitigate the financial burden of increased drug expenditures. The decline in both outpatient spending and outpatient OOP spending may indicate that targeted therapy is offsetting the need for outpatient services, potentially reflecting improved disease control. Additionally, we found a modest decrease in inpatient spending and virtually no change in inpatient OOP spending, likely due to the infrequent nature of inpatient visits in a low-acuity condition such as AD contributing to the noisy OOP spending estimates.

These findings have several implications for managed care pharmacy. Targeted therapy was associated with fewer all-cause hospitalization and any ED visits, suggesting that the effective systemic treatment for AD may reduce high-cost acute care use. Nevertheless, the modest declines in inpatient and outpatient spending were insufficient to offset the substantial rise in pharmaceutical spending, resulting in a net increase in total healthcare costs. Thus, managed care decision makers should balance pharmacy budget impact with potential reductions in acute and urgent care use, while also considering prior authorization criteria, patient affordability, and adherence.

To our knowledge, this is the first real-world study on the associations between targeted therapy, healthcare utilizations and spending among individuals diagnosed with AD. The percentage reductions in hospitalizations and ED visits were generally consistent with the reductions in outpatient and inpatient spending. These results may be utilized in future models that evaluate the real-world value of targeted therapy in treating AD patients. Although our study reflects the current treatment landscape for moderate-to-severe AD, it is important to note that dupilumab accounted for most of the targeted therapy population. Key strengths of this study include the use of a large commercial claims database, MarketScan, and a DiD design that controlled for time-invariant individual characteristics and calendar time shocks common to both groups.

Limitations

This study has several limitations. First, the population consisted of AD patients with employer-sponsored commercial health insurance as we used MarketScan. Therefore, the findings may not be generalizable to patients with other types of insurance or uninsured populations. Second, as the comparison group included patients with AD who did not receive targeted therapy, these patients may still have received topical therapy, conventional systemic therapy, supportive care. Thus, our estimates should be interpreted as the association of targeted therapy initiation relative to non-targeted-therapy care.

Third, even though our models included person fixed effect to control for time-invariant individual characteristics and calendar year-month fixed effect to control for common secular trends and seasonality, residual unobserved confounding may remain. The validity of our estimates relies on the assumption that any unmeasured time-varying clinical factors trended similarly between the targeted therapy and comparison groups over the study period, and our estimates could be biased if these factors diverged systematically between groups.

Fourth, hospitalizations and ED visits were measured as all-cause outcomes, so individual encounters cannot be attributed with certainty to AD, treatment failure, treatment-related adverse events, or unrelated conditions. Our difference-in-differences design assumes that any AD-unrelated acute care use trended similarly between the targeted therapy and comparison groups over the study period; if this assumption does not hold, our estimates could still be biased.

Fifth, our 12-month outcome window did not require continuous treatment through the point of each encounter, so our estimates reflect an average effect across patients regardless of treatment persistence and may not represent the effect specific to patients who remained on targeted therapy throughout follow-up.

Sixth, claims data do not capture manufacturer copay coupons or other assistance programs, which may overstate patients' net OOP burden, particularly for targeted therapies where such programs are common. This affects our OOP spending estimates only, not our total healthcare spending estimates, which capture full cost of care regardless of payer.

Finally, most patients in the targeted therapy group received dupilumab, so the findings may not be generalizable to newer targeted therapies or to individual biologic and JAK inhibitor agents.

Conclusion

In this retrospective cohort study of commercially insured adults with AD, initiation of targeted therapy was associated with lower probabilities of all-cause hospitalization and ED visits compared with non-targeted therapy care. The use of targeted therapy was associated with lower inpatient and outpatient spending but higher total healthcare and drug spending, while patient out-of-pocket spending increased only slightly. These findings suggest that targeted therapy may reduce high-intensity acute care use among adults with AD, but with important tradeoffs in pharmacy and total healthcare spending. Future studies should investigate further about AD-specific acute care use, treatment-related adverse events, and the longer-term value of targeted therapies across different patient populations.

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Tables

Table 1. Study Population Demographics and Comorbidities

Characteristics, N (%)[a]	Treated Group (N=9,320)	Comparison Group (N=148,646)	P-value
Age – mean (SD)	41.13 (14.14)	41.58 (13.57)	0.008
Female	5,408 (58.0)	97,481 (65.6)	<0.001
Male	3,912 (42.0)	51,165 (34.4)	
Region			
Northeast	1,185 (12.7)	24,023 (16.2)	<0.001
North Central	1,232 (13.2)	20,012 (13.5)	
South	3,803 (40.8)	56,079 (37.7)	
West	959 (10.3)	19,008 (12.8)	
Unknown	2,141 (23.0)	29,524 (19.9)	
Comorbidity Index			
0	4,258 (45.7)	85,301 (57.4)	<0.001
1	3,027 (32.5)	34,951 (23.5)	
2	833 (8.9)	13,593 (9.1)	
≥3	1,202 (12.9)	14,801 (10.0)	
Targeted Therapy			
Abrocitinib	0	NA	
Baricitinib	8 (<0.1)	NA	NA
Dupilumab	9,247 (99.2)	NA	
Tralokinumab	1 (<0.1)	NA	
Upadacitinib	64 (0.7)	NA	
No Treatment	NA	148,646 (100.0)	

[a]Number and percentages of individuals are shown for each categorical variable, except for patient

age where mean and standard deviation are shown.
SD = standard deviation;

Table 2. DID Results on Primary Outcomes

Outcome	Unadjusted means[a]				Adjusted Treatment Effect (95% CI)[b]
	Comparison Group		Treated Group		
	Pre-Index	Post Index	Pre-Index	Post Index	
Any hospitalization	0.070	0.068	0.072	0.060	-0.016*** (-0.021 to -0.010)
Any ED visit	0.162	0.169	0.190	0.179	-0.017*** (-0.025 to -0.008)

[a]Unadjusted means are the raw means of 1-year outcomes for the given cohort in the period outlined.

[b]The adjusted treatment effect reports results from linear regressions that control for person-indicators, calendar year-month indicators, and CCI score. Coefficients on the interactions between an indicator for the treatment group and an indicator for the post period are reported, with 95% confidence intervals in parentheses. Standard errors are clustered at the individual patient level. *p<0.1; **p<0.05: ***p<0.01.

CI = confidence interval; DiD = difference in difference; ED = emergency department

Table 3. DID Results on Total Healthcare Spending and Spending by Type

Outcome	Unadjusted means, $[a]				Adjusted Treatment Effect (95% CI)[b]
	Comparison Group		Treated Group		
	Pre-Index	Post Index	Pre-Index	Post Index	
Total Spending	7978.58	8954.78	11,010.60	37,934.62	24,901.01*** (24,482.26 to 25,319.76)
Drug Spending	2,155.83	2,498.88	4,215.51	31,000.12	26,061.26*** (25,734.560 to 26,387.960)
Outpatient Spending	4,826.84	5,396.16	5,691.48	5,887.804	-909.25*** (-1,106.970 to -711.520)
Inpatient Spending	995.91	1,059.75	1,103.61	1,046.69	-251.01*** (-382.273 to -119.739)

[a]Unadjusted means are the raw means of 1-year outcomes for the given cohort in the period outlined.

[b]The adjusted treatment effect reports results from linear regressions that control for person-indicators, calendar year-month indicators, and CCI Score. Coefficients on the interactions between an indicator for the treatment group and an indicator for the post period are reported, with 95% confidence intervals in parentheses. Standard errors are clustered at the individual patient level. *p<0.1; **p<0.05: ***p<0.01.

CI = confidence interval; DiD = difference in difference

Figures

Figure 1. Study Design

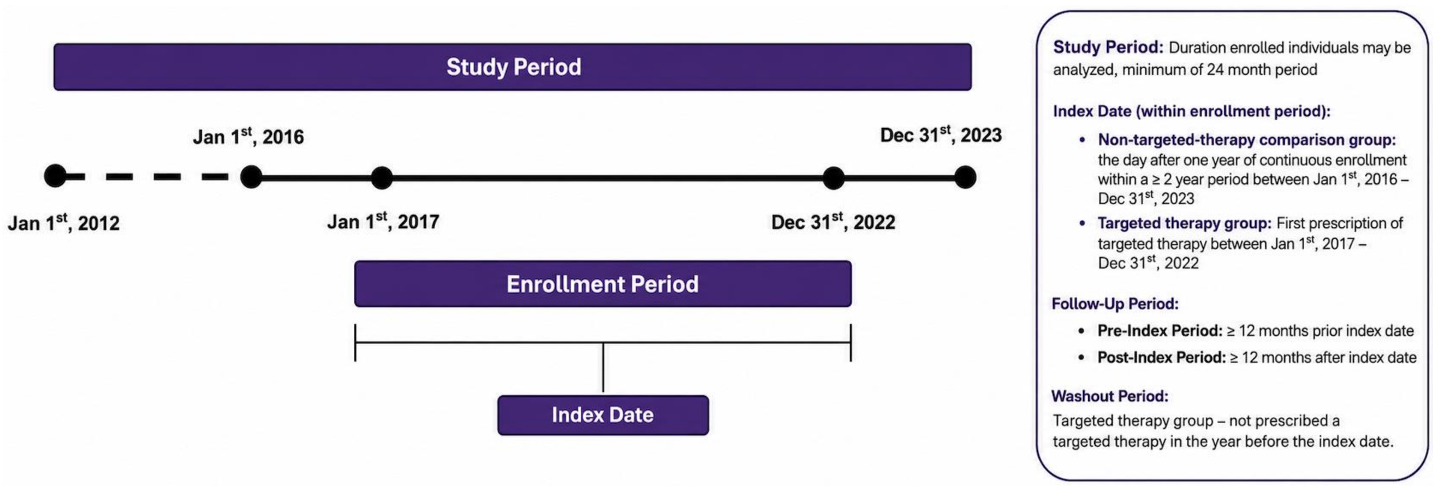
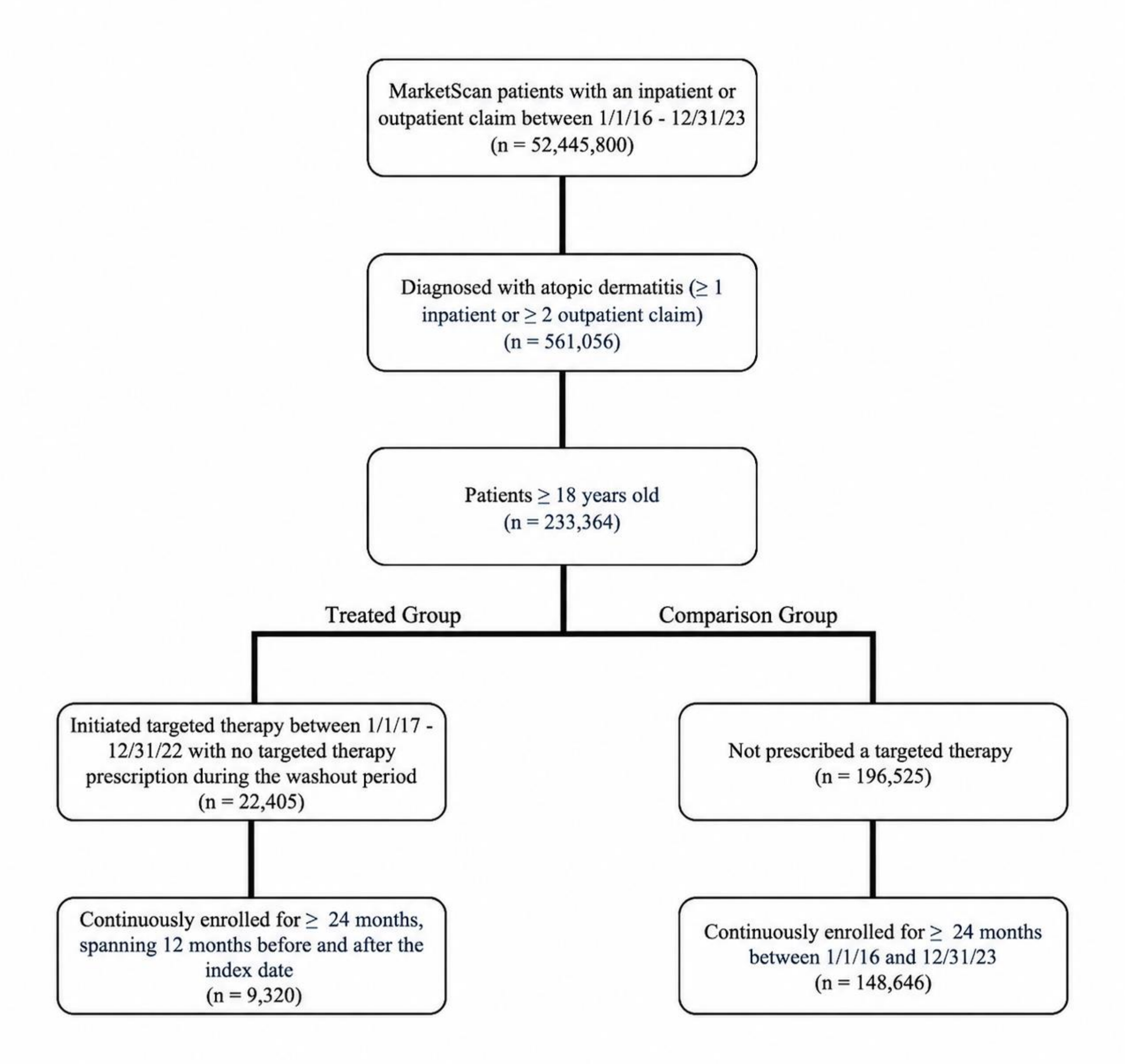


Figure 2. Cohort Selection and Attrition Diagram



Supplemental Online Content

Ta R, Jung J, Lee K, Li J. Impact of Targeted Therapy on Healthcare Resource Utilization Among Patients with Atopic Dermatitis

SUPPLEMENTARY METHODS 1. DiD Regression Model
SUPPLEMENTARY FIGURE 1. Distribution of Person-year
SUPPLEMENTARY FIGURE 2. Parallel Trends Graph of Healthcare Resource Utilization
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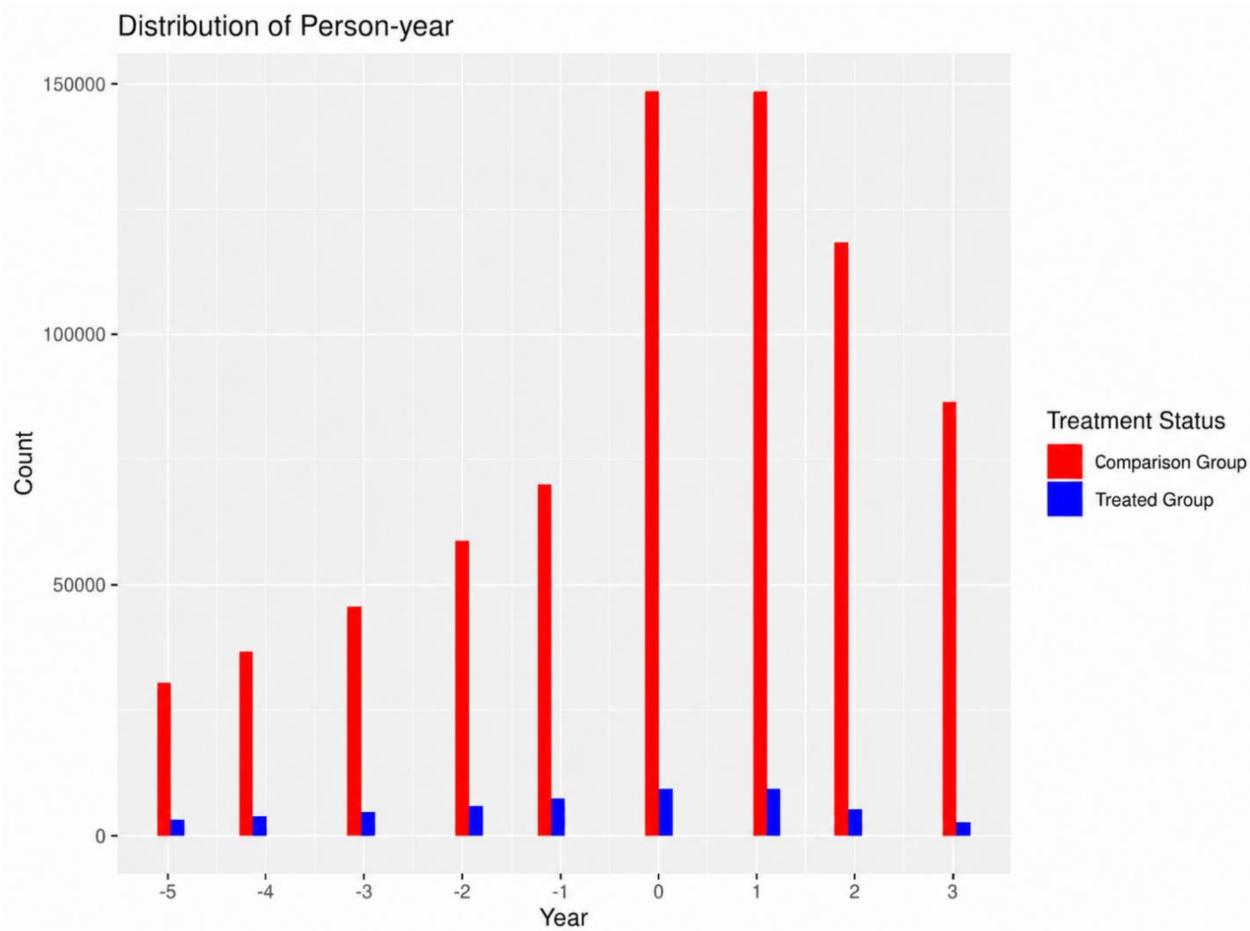
SUPPLEMENTARY METHODS 1. DiD Regression Model

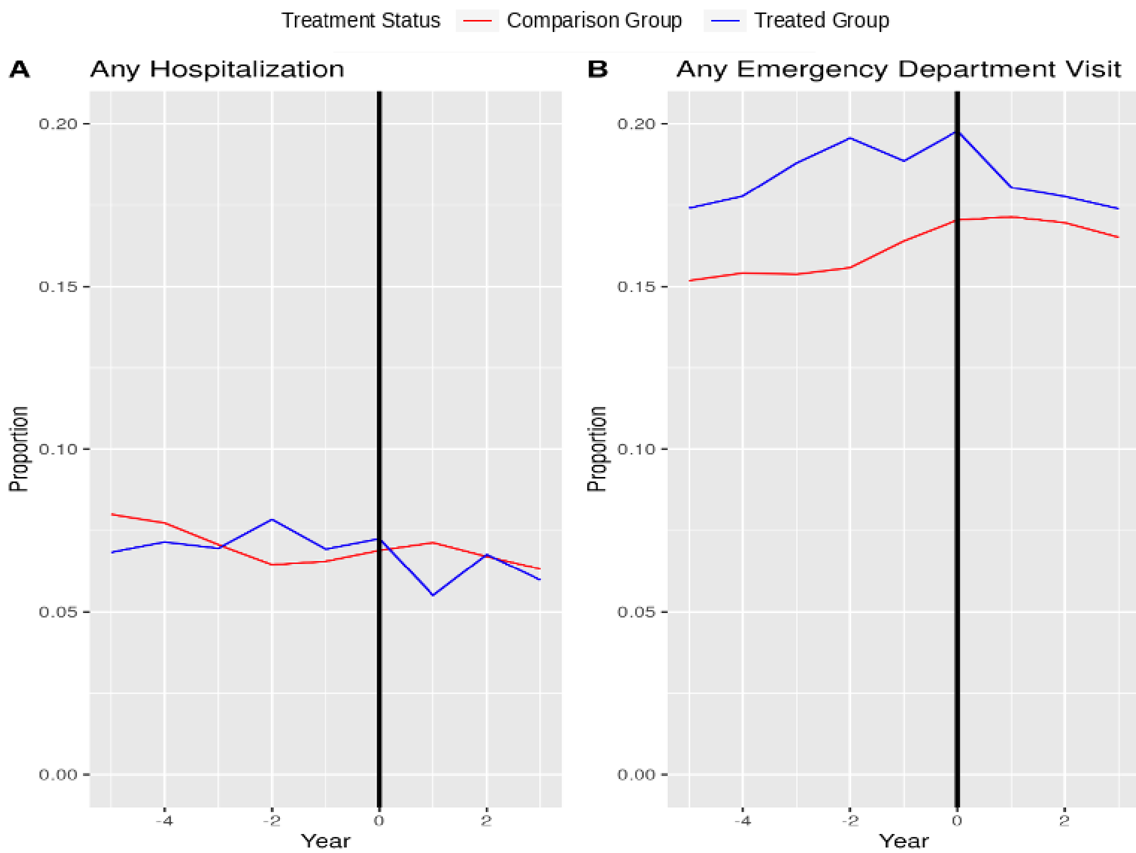
$$Y_{it} = \beta_0 + \beta_1 post_t + \beta_2 treated_i * post_t + \beta_3 calyear_t * month_t + \beta_4 CCI_{it} + \alpha person_i + \delta\, calyear_t + \theta month_t + \varepsilon_{it}$$

Y_{it}: outcome of interest (ED visit, inpatient visit, and total cost)
B_1: coefficient on an indicator for if the year falls after index date
$Treated_i$: indicator for the targeted-therapy group
B_2: the DiD coefficient of interest
B_3: coefficient of interaction between calendar year and quarter
B_4: coefficient for Charlson Comorbidity Index (time varying)
: coefficient on an indicator for individual, I
δ: coefficient on an indicator for year, t
: coefficient on an indicator for month
ε_{it}: the error term clustered at the individual level

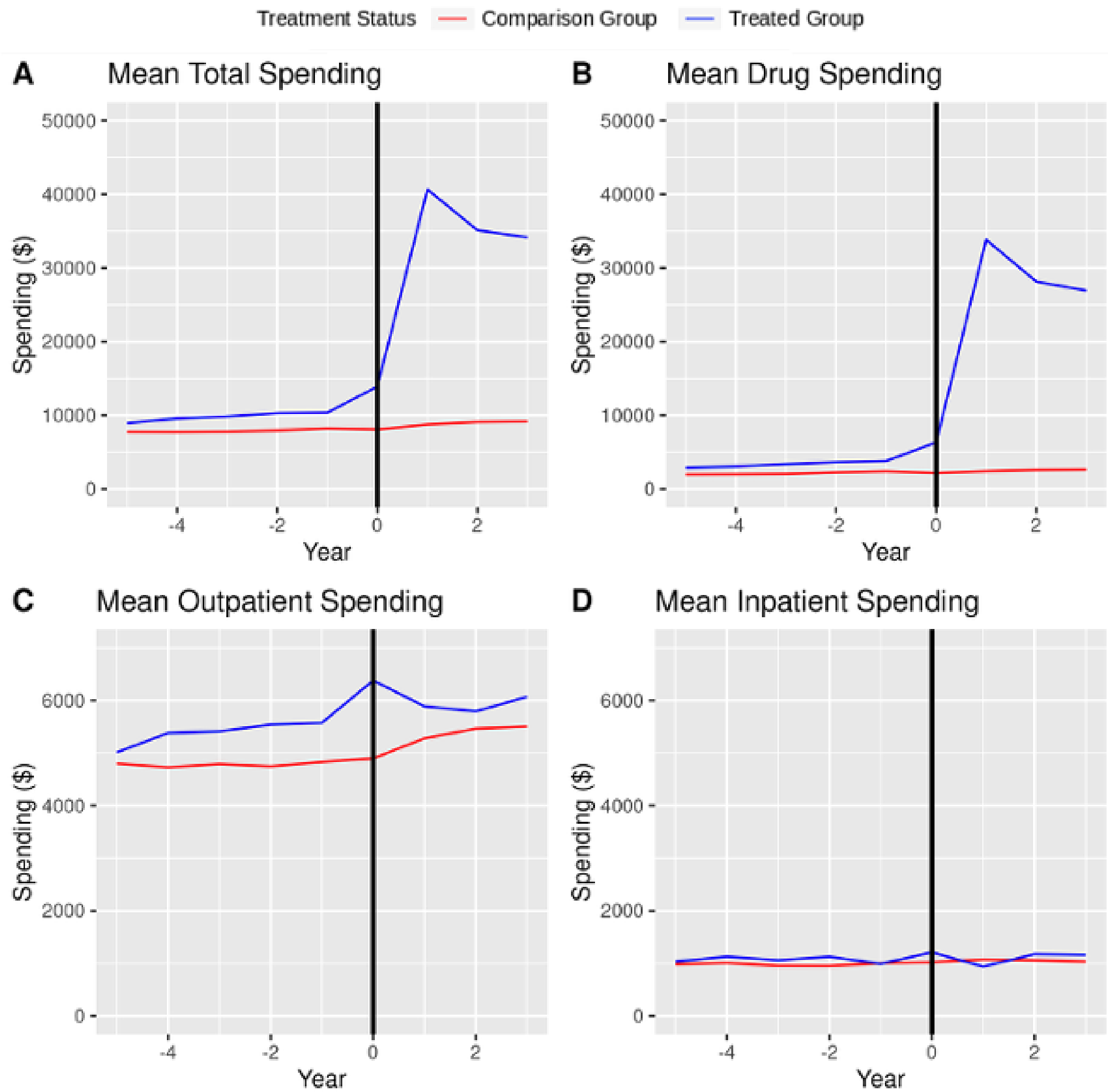
SUPPLEMENTARY FIGURE 1. Distribution of Person-year



SUPPLEMENTARY FIGURE 2. Parallel Trends Graph of Healthcare Resource Utilization



SUPPLEMENTARY FIGURE 3. Parallel Trends Graph of Spending



SUPPLEMENTARY TABLE 1. ICD-10 Codes Used to Identify Atopic Dermatitis

Description	ICD-10 Code
Atopic dermatitis	L20, L20.8, L20.89, L20.9

SUPPLEMENTARY TABLE 2. Targeted Therapy NDC Codes

Drug	NDC Codes
Abrocitinib	00069023530, 00069043530, 00069033530, 63539023614, 63539033614, 63539043614
Baricitinib	00002418230, 00002418261, 00002447930, 00002473230, 00002447961
Dupilumab	00024591401, 00024591502, 00024591500, 00024591601, 00024591801, 00024591901, 00024591902, 00024591900, 00024591920, 00024591100, 00024591101, 00024591102, 00024591120, 00024591400, 00024591402, 00024591420, 00024591501, 00024591520, 00024591800, 00024591802, 00024591820
Tralokinumab	50222034602, 50222034604, 50222034601, 50222034622, 50222034691, 50222034692
Upadacitinib	00074230630, 00074104328, 00074231030, 00074104314, 00074230670, 00074231020

SUPPLEMENTARY TABLE 3. DID Out of Pocket Spending Results

Outcome	Unadjusted means, $[a]				Adjusted Treatment Effect (95% CI)[b]
	Comparison Group		Treated Group		
	Pre-Index	Post Index	Pre-Index	Post Index	
Total OOP Spending	394.86	277.24	448.21	490.50	95.10*** (82.51 to 107.69)
Drug OOP Spending	107.48	62.72	141.61	263.44	140.35*** (132.03 to 148.66)
Outpatient OOP Spending	286.99	214.30	306.27	226.79	-45.29*** (-55.79 to -34.80)
Inpatient OOP Spending	0.39	0.21	0.33	0.27	0.04 (-0.02 to 0.105)

[a]Unadjusted means represent the mean of 1-year outcomes for the given cohort in the period outlined. [b]The adjusted treatment effect reports results from a linear regression that controls for person-indicators, calendar year-month, and CCI. Coefficients on the interaction between an indicator for the treatment group and an indicator for the post period are reported, with 95% confidence intervals in parentheses. Standard errors are clustered at the individual patient level. *p<0.1; **p<0.05: ***p<0.01.